

HCP Profiling App • iPad Territory Reconfiguration Blueprint

Concise Executive Implementation Plan: Tablet Dual-Pane UX, ProMotion 120 FPS & Zero-Lag Architecture

Target Platform: Apple iPad (iPadOS 16+) / 10.9" to 12.9" Pro

Performance Goal: Locked 120 FPS ProMotion • 0ms Touch Latency

Core Business Driver: Field Headcount Downsizing Consolidation

Golden Rule: Strictly No Territory Code Renamed

1. Business Driver & The Excel Comparison Baseline

Manpower Consolidation Reality

Downsizing

- Field Headcount Constraints:** Operational instances require reducing reps while maintaining customer coverage.
- Smaller District Groupings:** Territories are regrouped into denser, compact clusters within existing districts.
- Coverage Redistribution:** Surviving reps absorb consolidated target accounts without losing doctor history.

Excel Comparison Dataset

Dataset Review

- Previous Dataset:** Historical territory codes, past MedRep ownership, and old district boundaries.
- Proposed Dataset:** Downsized team assignments, consolidated groupings, and updated target quotas.
- Leadership Sign-Off:** Higher officials and DSMs meet first to evaluate and finalize this comparison sheet.

🚨 ABSOLUTE ARCHITECTURAL RULE: "NO TERRITORY CODE SHALL BE RENAMED"

Territory codes (e.g., `RM106`, `BAYER-T01`) are primary keys tied to past doctor accounts, call logs, and sales invoices. **Renaming is strictly disabled at widget level.** Realignment is performed only via **re-parenting**, **re-assigning sales reps**, and **archiving**.

2. iPad-Optimized Dual-Pane Workspace (Landscape & Portrait)

IPAD TABLET DISPLAY (Split-View Dual-Pane • Apple Pencil & Multi-Touch Drag-and-Drop)

ROLE: DSM / ADMINISTRATOR ONLY

LEFT PANE: SOURCE DATASET (40% WIDTH)

- Excel Dataset Loader:** Ingest via iPad Files app, iCloud Drive, or AirDrop.
- Unassigned Territory Pool:** Draggable cards showing doctor counts (e.g., `RM106` • 18 Doctors).
- Search & Filter:** Real-time instant filtering by region, hospital count, or previous rep.

RIGHT PANE: TARGET DISTRICT CLUSTERS & ARCHIVE (60% WIDTH)

- Consolidated District Cards:** Denser groups with target rep dropdown selector.
- Drag-and-Drop Drop-Zones:** Instant visual snap-in when dropping territory nodes.
- Archive Drop-Zone:** Dedicated target container for decommissioned codes.

3. The 5-Pillar Zero-Lag Mobile Architecture (Locked 120 FPS ProMotion)

1. Dart Worker Isolate

compute()

- Offloaded Processing:** Excel parsing and tree diffing run on secondary thread.
- Cycle Detection:** Validates hierarchy to prevent circular references before commit.
- Zero UI Frame Drop:** Guarantees locked 120 FPS on iPad Pro displays.

2. Optimistic UI Updates

0ms Latency

- Instant Touch Feedback:** Tree mutations reflect immediately in local memory.
- Zero Modal Buffering:** No full-screen loading spinners or frozen screens.
- Subtle Status Pill:** Non-intrusive background sync indicator in top bar.

3. SQLite WAL Staging

Offline Queue

- Atomic Batch Commits:** Staging table records re-parenting in < 10ms.
- Write-Ahead Logging:** Fast disk I/O without UI thread contention.
- 100% Offline Safe:** Safe against elevator and basement connectivity loss.

4. Non-Blocking REST Worker

Background Sync

- Chunked Processing:** Communicates with ERPNext in 20-node silent batches.
- Persistent Connection:** Uses reusable HTTP client with exponential retry backoff.
- Uninterrupted Multitasking:** User can switch to doctor listing while sync runs.

5. Immutability & Role Shield

Governance

- Locked Primary Keys:** Widget architecture forbids renaming territory codes.
- Role-Gated Access:** Menu option visible only to Administrator and DSM logins.
- Rollback Protection:** Automatically restores local state if server errors occur.

Operational Execution, Archive Policy & Technical Blueprint

Step-by-Step Flow, Preservation Strategy, REST API Endpoints & Implementation Checklist

Backend: ERPNext v15 Multi-Tenant	Security: Session Cookie (sid) + CSRF Token	Data Integrity: Zero Deleted Records (Archive Node)
Sync Strategy: 15-Min TTL Invalidation		

4. The 4-Stage Operational Reconfiguration Flow

STAGE 1 ALIGNMENT	Higher Leadership & DSM Alignment Meeting - Leadership and DSMs evaluate Excel sheet comparing Previous Dataset vs. Proposed Dataset. - Consolidates field territories into smaller groups in the same commercial districts. - Enforces strict validation that no territory code has been renamed.	MANPOWER REDUCTION
STAGE 2 IMPORT	iPad In-App Dataset Staging & Validation - Administrator or DSM opens iPad app and imports finalized Excel via iPadOS Document Picker. - Dart Worker Isolate parses rows off-thread and populates the Left Pane unassigned pool. - Validates hierarchy boundaries and flags invalid parent references before user interaction.	IPADOS FILES
STAGE 3 EXECUTION	Tactile Dual-Pane Re-Parenting & Archive Relocation - User drags territory cards from Left Pane into Right Pane consolidated district groups. - Assigns surviving Medical Representative to each consolidated group via dropdown. - Drags obsolete/downsized territory codes into the Archive Drop-Zone ('Archived Territories').	DRAG & DROP
STAGE 4 PUBLISH	Silent Background Synchronization & Field Cutover - User taps "Publish Reconfiguration"; SQLite staging queue commits in single atomic batch. - Background sync worker transmits updates in 20-node chunks to ERPNext REST API. - Field MedReps immediately profile doctors under new smaller groups with zero app re-install.	LIVE PUBLISH

5. The Archive Policy: What Happens to Removed or Adjusted Territories?

Why We Archive Instead of Delete	DATA INTEGRITY	Ready for Future Manpower Expansion	RE-PUBLISHABLE
Foreign Key Protection: Deleting rows in ERPNext breaks historical call logs, invoices, and doctor accounts ('HCP Account'). Relocation Only: Decommissioned codes are simply re-parented to Archived Territories. Doctor Affiliations Intact: Existing doctor records remain 100% accessible with zero orphan states.			
Instant Headcount Restoration: If manpower is added back, codes are already fully configured. Single-Step Reactivation: Admin drags territory from Archive back to an active district on the iPad. Historical Continuity: Past sales history and doctor links immediately resume without data loss.			

6. Exact ERPNext REST API Endpoints Powering iPad Realignment

GET /api/resource/Territory	ACTIVE INGESTION
Query Filter: filters=[["parent_territory","!=","Archived Territories"]] • limit_page_length=1000. Purpose: Downloads active territory nodes into iPad local cache while excluding archived codes.	
GET /api/resource/Sales_Person	SALES HIERARCHY
Purpose: Downloads sales hierarchy to populate rep assignment selectors in consolidated district cards.	
PUT /api/resource/Territory/{territory_code}	RE-PARENTING
Payload: {"parent_territory": "New District Group"} or {"parent_territory": "Archived Territories"}. Server Impact: ERPNext automatically recalculates MariaDB nested set indexes ('lft', 'rgt').	
PUT /api/resource/HCP_Account/{id}	IN-PLACE SYNC
Payload: {"territory": "RM106", "sales_person": "REP_ID"}. Zero Duplicates: Updates doctor territory affiliation in-place without generating duplicate doctor profiles.	

7. Actionable Implementation & Rollout Checklist

Phase 1 (Isolate & Staging): Implement `territory_reconfig_isolate.dart` and `staged_territory_reconfig` SQLite table. Phase 2 (iPad Dual-Pane UI): Build `territory_reconfig_screen.dart` with dual-pane layout and drag-and-drop cards. Phase 3 (AppDrawer Gating): Expose menu item strictly for `UserPosition.admin` and `UserPosition.dsm` logins. Phase 4 (Background REST Sync): Implement chunked 20-node sync worker with exponential retry backoff. Phase 5 (Field Verification): Conduct end-to-end test on iPad device: Excel import → drag-and-drop → background publish.	
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